



The impacts of collective threshold requirements for rewards in a CPR experiment

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Abstract

Rewards for desirable behavior are increasingly being adopted in many regulatory settings. Nevertheless, there is no consensus on how rewards should best be crafted to achieve desired results. This paper focuses on the issue of regulatory design in the context of forest management in Thailand. We run lab-in-field experiments using a common-pool-resource game with villagers in northern Thailand. We study the impacts of varying the collective threshold for obtaining economic rewards on resource extraction, distribution of payoffs and cost efficiency. Three thresholds are studied—strict, moderate, and relaxed, corresponding to low, moderate, and generous targets for group extraction accordingly. We find that rewards are associated with positive conservation outcomes and that their effectiveness depends on the design of the reward instrument. In our setting, strict and moderate thresholds lead to the greatest gains in conservation but differ in important ways. The strict threshold leads to cost efficiency but induces free-riding behavior. The moderate threshold is not as cost-efficient but allows for a more equitable payoffs distribution. The relaxed threshold does not lead to any gain in conservation and is the least cost-efficient. Increasing the reward size can induce a faster achievement of the equilibrium outcome.

Keywords Collective threshold · Common-pool resource · Experiment · Payments · Reward size

JEL Classification Q23 · Q28

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1 Introduction

Ever since Hardin (1968)'s work predicting resource depletion as the natural outcome when rational and self-interested individuals have control over common-pool resources (CPR),¹ researchers have examined various factors that could lead to the effective management of the commons. While researchers agree that success depends on institutional arrangement, there is less agreement on the design of institutions that would lead to effective CPR management.² On the policymakers' side, there is increasing use of economic rewards to motivate social cooperation. This can be seen from the proliferation of payments for ecosystem services (PES) initiatives, as well as the adoption of the United Nation's REDD+³ scheme, which emerged from international climate negotiations.

Despite this shift in policymaking, research on the use of rewards is relatively scarce, especially when compared with work on command-and-control regulations. The relatively small, but growing literature on the design of PES schemes studies features such as individual versus group subsidy, awarding rewards for all good behavior or for when a threshold is met, the interactions between the former and the latter features, and the issue of intrinsic and extrinsic motivations. Few papers have studied the impact of varying the level of the collective threshold or required condition for triggering the reward and the effect of changing the reward size. Of those that do (e.g. Travers et al. 2011; Kaczan et al. 2017), these features are not analyzed and discussed in detail as they are not the focus of their papers. Thus, our work aims to contribute to the literature by specifically studying these design features using lab-in-field experiments to examine the impacts of institutional design on conservation outcomes, distribution of payoffs, and cost efficiency.

We examine the impacts of rewards in the context of forest management in Thailand. The setting is one of farming communities who live on the hills in the northern province of Nan where deforestation is a big problem. In 2003, 74% of the province was covered in forests. A decade later, forest cover declined to 61%. Maize farming tripled in area from 46,934 ha in the early 2000s to 106,385 ha in 2017. While it is difficult to estimate how much forest land was converted to make way for maize, the practice is not uncommon in the area. Deforestation in the province happens despite the prevailing logging ban imposed by the Thai government and the imposition of hefty fines if encroachers are caught. Unfortunately, enforcement of such regulations is limited because of the largeness of the area and other constraints. With the Nan

¹ CPR is defined here as resource systems that produce resource units that are rival but non-excludable. Rival here is taken to mean rivalry in consumption. If a person has extracted the resource from the commons, that same unit of resource is taken and cannot be extracted by others. Non-excludability here means that it is not possible to exclude the resource from potential users of it either because of the nature of the property rights structure, or due to lax enforcement of the property rights over the resource.

² While the various works by Elinor Ostrom do identify the key components of institutions that work. There are mixed findings regarding the effectiveness of specific institutional design issues (e.g. using rewards or command-and-control, enforcement mechanisms, awarding rewards based on individual or group efforts, reward size etc.).

³ REDD stands for Reducing Emissions from Deforestation and forest Degradation.

River contributing to more than 40% of water in Thailand's lifeline, the Chao Phraya River, the deforestation problem in the Northern provinces is an important problem for the country.

In light of this, we address this question of how to design institutions that would lead to sustainable forest management by running lab-in-field experiments using a CPR game with upland villagers in Nan. We focus on two aspects of institutional design. The first is the collective threshold, which studies the effects of varying the threshold level of group's extraction for obtaining rewards. The second is reward size, which examines the impacts of different reward sizes. In this paper, thresholds are placed on the collective outcome. If a group's extraction is within the threshold, all members of the group receive the same reward. We examine the effects of varying the threshold level and reward size on (i) resource extraction, (ii) individual pay-offs, and (iii) cost efficiency.

We find that, in our setting, rewards with a collective threshold induce conservation behavior and that their effectiveness depends on the design of the reward instrument. Overall, the strict threshold (i.e. a low group target of resource extraction) results in the greatest gains in conservation and is the most cost-efficient. However, it also induces free-riding, resulting in a more uneven distribution of payoffs among players. The moderate threshold (i.e., a medium level of group target for resource extraction) leads to equally high conservation gains and allows for a more equitable payoffs distribution. The relaxed threshold (i.e., a high group target for resource extraction) does not lead to any gain in conservation and is the least cost-efficient. We also find that increasing the reward size can induce a faster achievement of the equilibrium outcome.

2 Literature review

With the conservation of common pool resources the exception rather than the norm (Hardin 1968; Olson 1965), academics have sought for external and internal factors that could induce cooperation in this social dilemma setting. External motivation comes in the form of institutional arrangements (see e.g., North 2006; Ostrom 2009; Whitmeyer 2002), while internal motivation comes in the form of social preferences such as fairness, reciprocity, and the willingness to abide by commitments or to punish those not complying (see e.g., Falk et al. 2002; Fehr et al. 2002; Fehr and Gächter 2000).⁴ The experimental literature finds that behavior in social dilemma situations can be strongly influenced by institutions⁵ (Castillo et al. 2011; Rodriguez-Sickert et al. 2008; Travers et al. 2011). Furthermore, social preferences can vary depending on institutional contexts (Biel and Thøgersen 2007). As such, putting in place appropriate institutions can help induce cooperation in social dilemma settings and limit

⁴ Note that Ostrom also studies internal motivation (see, e.g., Ostrom 1990, 2000).

⁵ Institutions are the 'rules of the game,' which encompass formal rules (such as rules and regulations), informal constraints (such as conventions, norms, and self-imposed codes of conduct), and the enforcement of both (North 1995).

resource exploitation (North 2006; Ostrom 2009; Whitmeyer 2002). Nonetheless, the suitability of institutions depends on their design, of which the use of economic incentives is one aspect.⁶

In social dilemma settings, economic incentives are applied to reduce the discrepancy between self-interest and group interest. Incentives that are greater than the cost of cooperation would change the outcome of the social dilemma problem to one conducive to the collective interest (Bravo and Squazzoni 2013). This reasoning applies whether the economic incentive used is a punishment or a reward. Findings from social dilemma experiments show that, for the most part, economic incentives matter in inducing cooperation in situations where achieving collective action for the good of all is a challenge (see, e.g., Fehr and Gächter 2000; Mulder 2008; Parks et al. 2001; Rand et al. 2009).⁷ Nonetheless, the extant literature mostly studies the use of negative incentives despite the increasing application of positive incentives by policymakers (Balliet et al. 2011; De Geest and Dari-Mattiacci 2013). A much smaller, but growing, number of papers look at the role of rewards, and fewer still have looked at the role played by the design of the collective condition for obtaining rewards. The design of positive incentives is the focus of our paper.

The group of literature focusing on the use of positive incentives such as payment for ecosystem services (PES) in managing CPRs mostly study the problem in the context of forest management in developing countries (see, e.g., Handberg and Angelsen 2015; Moros et al. 2017; Salk et al. 2017). At its beginning in the late 1990s, PES schemes focused mainly on individual contract and payment (Kerr et al. 2014). There are studies showing that PES with individual contracts results in better conservation outcomes by providing rewards that subsidize conservation based on individual efforts (see, e.g., Gatiso et al. 2018; Handberg and Angelsen 2015, 2019; Kaczan et al. 2019; Kuhfuss et al. 2016; Le Coent et al. 2014). However, the impacts and implications vary according to the setup and context. Handberg and Angelsen (2015) varies the level of individual payments and shows that increasing payments has diminishing effects on harvest rates.

In more recent literature, PES with collective agreement, where everyone is paid the same amount based on group effort, has started gaining attention from researchers and planners. This happens especially in developing countries where there are many farmers who own small pieces of land and are scattered around the forest area. In this setup, using group subsidies might be more practical as they help reduce transaction costs (Huang et al. 2009). Nonetheless, when comparing group and individual payments, the literature has mixed findings. While some find that group PES is effective in encouraging conservation behavior (Andersson et al. 2018), others find that group PES is likely to increase free-riding behavior and individual PES

⁶ Another aspect that is, for example, often studied is whether institutions are externally imposed or endogenously crafted. Mixed results have been found regarding the effectiveness of each type of institutions (see, e.g., Agrawal 2001; Armin et al. 2006; Ostmann 1998; Ostrom 2000).

⁷ This is not to say that any incentives would work to achieve sustainable outcomes. Effectiveness of incentives also depends on the other important characteristic of institutional design—the participation of local communities in the crafting of institutions.

is more preferable (see, e.g., Gatiso et al. 2018; Kaczan et al. 2019; Narloch et al. 2012).

Apart from the simple group PES, another interesting topic is the role of collective thresholds, where players will obtain rewards only if the group's resource extraction lies within the specified threshold. One group of literature studies the impacts of imposing collective thresholds with individual subsidy—an individual is awarded according to his/her own decision when the group target is met. For example, Le Coent et al. (2014) runs threshold public good games and finds that conditional subsidy, where each contributor is subsidized only if the public good threshold is attained by the group, is as effective as non-conditional subsidy, where each contributor is subsidized regardless of group performance but is more cost-efficient. Kuhfuss et al. (2016), also supports the finding that collective thresholds help induce more conservation behavior, compared to unconditional subsidies.

The second group of studies investigates the impacts of collective thresholds imposed with group subsidies. Moros et al. (2017) studies the impacts of collective conditions with individual and group subsidies. They find that, with collective conditions, both group and individual subsidies increase conservation behavior. Group subsidy also encourages social motivation for conservation. Midler et al. (2015), however, show that individual subsidy is more effective than group subsidy, and that group subsidy will be effective when social ties within the group are strong or when communication is allowed.

In this paper, we are specifically interested in how different levels of collective thresholds can influence conservation behavior. Most of the literature on the threshold level is considered within the context of negative incentives in the form of varying sanction levels (Beckenkamp and Ostmann 1999; Ostmann et al. 1999). One interesting finding identified in Ostrom's famous design principles is that sanctions should be done gradually, and the severity of sanctions varies depending on the context (Ostrom 1990). This contradicts the common belief in mainstream economics that a stricter rule should be more effective than a compromising one. However, sanctions are not a feature of positive incentive tools. Rather, when rewards are used, the rule governing when rewards are awarded, and the size of the reward are the design features that could motivate prosocial behavior among participants.

In this aspect, Travers et al. (2011) and Kaczan et al. (2017) are probably the closest works to the scope of our paper. Travers et al. (2011) investigate the impacts of different threshold levels in public good experiments with group payments. They find that with a very relaxed threshold where a lower level of conservation outcome is required, group payments insignificantly affect conservation behavior as only a minor adjustment is required to obtain the payments. Nevertheless, under a stricter threshold, group payments significantly induce better conservation outcomes compared with the control. They also show that individual extraction level is indifferent to a doubling of the reward. These results suggest that the threshold level plays a more important role than the payment size. Kaczan et al. (2017) ran a framed field-laboratory voluntary contributions experiment in Mexico. The paper compares results between high (contribute all) and low (contribute at least 40%) collective targets. Similar to Travers et al. (2011), they find that the threshold level positively impacts collective contribution only when it is strict, i.e., a low level of resource

extraction is permitted. When the target is relaxed and set close to the no-treatment extraction level, the threshold has little to no impact on contributions. However, both papers cover only high and low thresholds but do not investigate further how more differentiated targets would affect extraction decision. This is the main issue of focus in our paper. Summary of the details of the related literature mentioned above can be found in ESM Appendix A.

3 Experimental design and data collection

In this study, subjects play a CPR game where each individual i chooses a level of extraction y_i from a resource that is shared in common. Maximum extraction per individual is up to a capacity constraint of $y_i^{\max}$. Extraction from the common pool provides direct positive private benefits to the extractor but imposes negative externalities on others. Payoffs are non-linear, and the resource stock replenishes after each round (see details below and in ESM Appendix B). The basic setup of the game, applied in the control rounds and the control groups, is similar to the ones found in Ostrom et al. (1994), Falk et al. (2002) and Velez et al. (2010).

For the treatment groups, when group extraction lies within the collective threshold, a reward of size R is equally added to the payoffs of each group member regardless of individual contribution. The payoff function for each individual is given by:

$$\Pi_i = P_M \times y_i + P_F \times [Y - (y_i + y_{-i})] + R \quad \text{where } R = 0 \quad \text{if } (y_i + y_{-i}) > Y_c,$$

where Π_i is the payoffs for individual i , P_M is the returns to individual i for extracting each unit of the resource. This is positively related to the remaining resource ($Y - (y_i + y_{-i})$), as the yield of individual land-use is likely to depend on the richness of the remaining forest.⁸

y_i is the individual i 's harvest decision, P_F is the returns to individual i for each unit of the commons that remains intact.

This return is assumed to be constant.⁹

y_{-i} is the sum of other individuals' harvests. Y is the total amount of resource available to a group before harvest. R is the reward given to a subject if the collective threshold is met. This is 0 for the baseline and control groups, and for the treatment groups that cannot meet the threshold.

Y_c is the threshold level of group extraction whereby $R=0$ if $(y_i + y_{-i}) > Y_c$.

In our study, the game is played iteratively within groups of 5 people. The maximum level of resource extraction is 80 per person ($y_i \leq 80$). Total available resource for the community is assumed to be 500 in each round. Payoffs for different levels of extraction are shown in Table B.1 in ESM Appendix B. Under the baseline scenario ($R = 0$), standard economic theory based on the assumption of rational self-interest predicts that the Nash equilibrium (NE) occurs when every subject extracts the

⁸ In our payoff table, P_M varies from 0.8 to 2 per unit. If the remaining resource reduces by 10 units, the return rate will decrease by 0.03 per unit.

⁹ We assume this is equal to 0.4.

Table 1 Details of the treatments

Treatments	Nash equilibrium	Social optimum	Threshold (resource use/group)	Reward size (per player)	Groups (subjects)
No threshold (control)	80	0	None	None	3 (15)
Relaxed threshold	80, 50	0	250	20	4 (20)
Moderate threshold_20	80	0	150	20	4 (20)
Moderate threshold_40	80	0	150	40	3 (15)
Strict threshold	80	0	0	20	3 (15)

maximum level ($y_i = 80$). In our setting, however, the social optimum (SO) lies at the opposite end of the extraction spectrum from the NE outcome. SO occurs when no one extracts from the resource ($y_i = 0$). The NE results in over-extraction of the CPR.

To mimic the village setting where the same people extract from the forest, the same group of 5 people play for multiple rounds. Extraction decision is private, but after each round is played, players get feedback on their group's level of extraction as well as their payoff for that round. Communication is not allowed. Payoff from the game comprises of two parts, a show-up fee to compensate players for their time, and earnings from the game. This latter part of the payoff is equal to the individual's average payoffs in all rounds of the game.

A total of 16 rounds are played by each group. To prevent end-game effect, the groups are not given information on the number of rounds. All groups play the basic game with no treatments—no limits on resource extraction—for 8 rounds. Groups are then randomly assigned to one of four possible treatments and play for another 8 rounds. One group continues to play the baseline scenario, other treatments vary depending on (i) the threshold, the maximum level of group extraction at and below which subjects obtain a reward and (ii) the size of the reward received if the group is within the specified extraction requirement. As mentioned above, three different levels of collective thresholds (Y_c) are imposed (relaxed, moderate, strict) with the same reward size. The relaxed threshold corresponds to a group in which total extraction is limited to 250 units of the resource (50 units per person on average). The moderate threshold corresponds to a group total of no more than 150 units (30 units per person on average). The strict threshold does not allow for any extraction at all (SO outcome). Rewards are equally awarded to all players in a group only if the group extraction does not exceed the threshold. To determine the effect of reward size, the moderate threshold treatment has two reward levels (R). These are low ($R = 20$ experimental currency units) and high ($R = 40$) respectively. Details of the treatments are summarized in Table 1.

We collected data from people who live within proximity of the forest in Nan province in the northern part of Thailand where deforestation is a big problem. Our approach is a lab-in-the-field experiment with abstract framing. We used a non-student subject pool and introduced the field context by asking subjects to think of the resources in the game as common pool resources they are used to utilizing. We did

Table 2 Socio-economic characteristics of participants

Variable	Mean	Std dev	Min	Max
Age (years)	51.76	10.35	20	77
Gender (1 = female)	0.48	0.50	0	1
Education (years)	8.48	3.10	0	16
Household annual income (Baht ^a)	79,934.97	78,466.69	25,000	500,000
Household size (persons)	4.13	1.10	2	8
Agricultural area owned by household (rai ^b)	16.05	20.39	0	120

^a1 Baht is approximately equal to 0.03 USD

^bRai is the traditional unit for measuring land in Thailand. 6.25 rai = 1 ha

not specifically say the resource was a forest because extraction from the majority of forests in our study area is banned (albeit with weak enforcement of the rules). Using the term ‘forest’ in the game would make subjects reluctant to participate or give biased responses.

In all, data was collected from 6 villages, and a total of 85 individuals participated in the game. Participants were randomly assigned to groups. The number of groups and subjects per treatment is shown in Table 1. The game was played using pen-and-paper. Table 2 shows the average socio-economic characteristics of the participants. Kruskal–Wallis tests show that subjects were successfully randomized into groups (see Table C.2 in ESM Appendix C).

4 Analysis and results

4.1 Impact of rewards on resource use

Results from the field support the hypothesis that the rewards result in a lower level of resource extraction compared with the no-treatment group. This finding is consistent with the existing literature. Figure 1 shows that for all treatments, average individual extraction in the treatment rounds (Rounds 9–16) is lower than the baseline rounds (Rounds 1–8). By contrast, the control group’s average extraction increases by 5.4 in Rounds 9–16 compared to Rounds 1–8. Within-subject comparison using the Wilcoxon signed-rank test indicates that applying rewards reduces extraction in a statistically significant way.¹⁰

Table 3 reports regression results.¹¹ Regression analysis confirms that using rewards significantly reduces resource extraction. Random effects models are estimated using the feasible generalized least squares (FGLS) method to control for autocorrelation with bootstrapped standard errors to control for non-normality of the

¹⁰ The test compares average extraction for all treated groups before and after treatment. The test statistic of 3.16 is statistically significant at the 1 percent level.

¹¹ Descriptions of variables are shown in Table C.1 of ESM Appendix C.

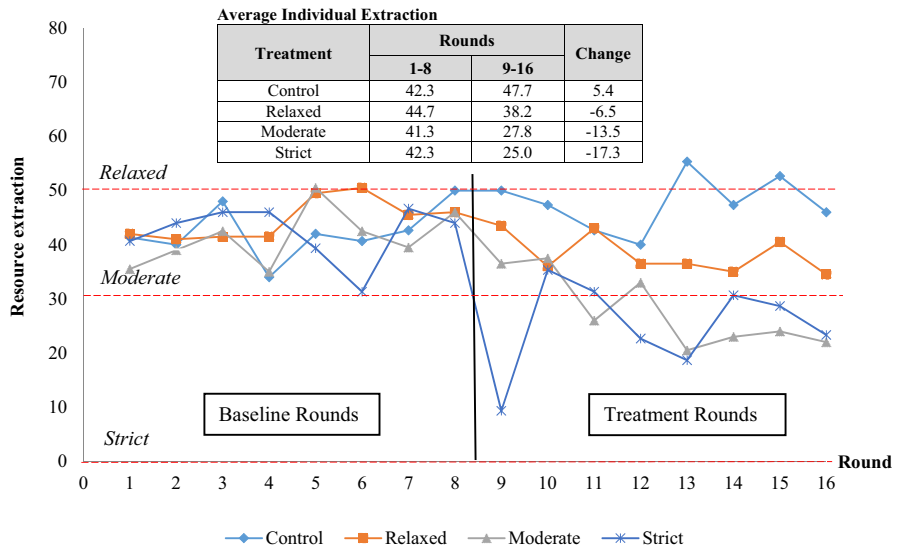


Fig. 1 Average level of individual extraction

Table 3 Regression results
[Models (1) and (2)]

	(1) RE		(2) RE	
	Coefficient	SE ^a	Coefficient	SE ^a
Reward	-8.561**	3.807		
Strict threshold			- 12.183**	5.152
Moderate threshold_20			- 9.998**	4.333
Moderate threshold_40			- 9.256**	4.318
Relaxed threshold			- 3.066	4.561
Group fixed effects	Yes		Yes	
Session fixed effects	Yes		Yes	
Socio-economic controls	Yes		Yes	
Wald Chi-square	81.42***		103.88***	
Observations	1275		1275	
Subjects	85		85	
Time periods	15		15	

The baseline is the control group with no treatments

***, **, and * next to the coefficient indicate significance levels of 1 percent, 5 percent, and 10 percent accordingly

^aReported figures are bootstrapped standard errors

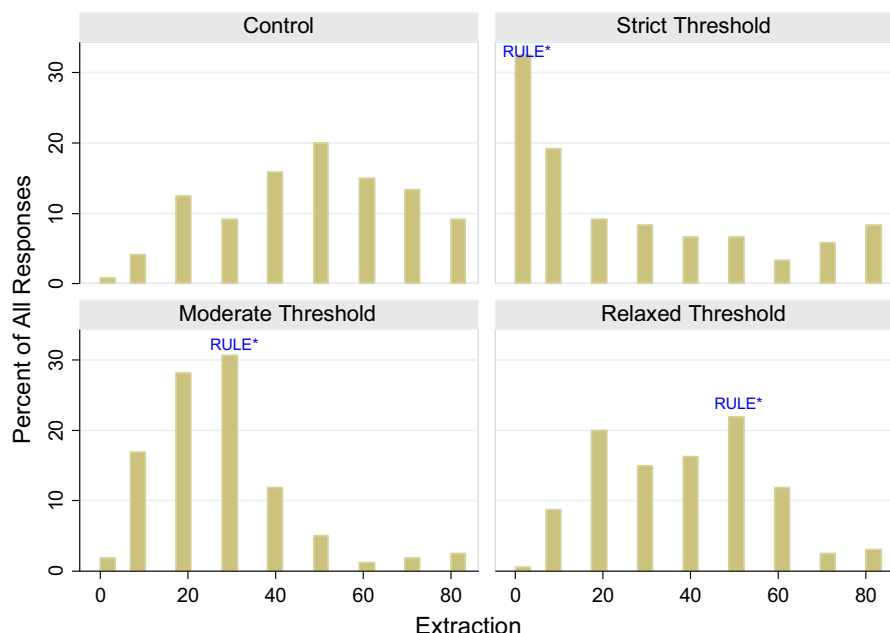


Fig. 2 Individual extraction decision by treatment

data¹² (Gonçalves and White 2005; Lindahl et al. 2015).¹³ In Model (1), all threshold treatments are grouped together into one dummy variable, reward. Its coefficient is statistically significant, which implies that, overall, conditional reward reduces extraction level by 8.561 units compared to the no-condition treatment. Thus, we conclude that, in our setting, using rewards helps to induce social cooperation and limits common pool resource use. Moreover, the results show that the level of extraction is positively related to group extraction in the previous round. This implies that subjects tend to extract more (less) if group extraction in the previous round was high (low). This behavior suggests that subjects are “conditional cooperators” as pointed out in Rabin (1993).

4.2 Impact of varying collective thresholds on resource use

We find that, in our setting, different collective conditions affect resource extraction differently. Compared to the baseline rounds, average extraction is reduced the most in the strict threshold treatment (17.3 units), followed by the moderate (13.5

¹² The Shapiro–Wilk W test rejects normality of the data for key variables (extraction choice, age, education, income, and household size). Test statistics can be found in Table C.3 of Appendix C.

¹³ Following Lindahl et al. (2015), we use 1000 replications.

units) and the relaxed (6.5 units) threshold treatments respectively.¹⁴ Random effects FGLS regression show that the strict and moderate thresholds lead to significantly lower resource use compared to the control, while the relaxed threshold does not significantly limit resource extraction [see Model (2) in Table 3]. However, the coefficients of the strict and moderate thresholds are not statistically significantly different from one another.¹⁵

An examination of individual data provides some information on the workings of collective threshold requirements. Figure 2 shows that subjects do try to abide by the assigned group target. For all treatment groups, the mode of extraction level is equal to the average target for an individual (i.e. 0, 30, and 50 for the strict, moderate, and relaxed thresholds accordingly). This finding, coupled with the results from statistical tests and regression analysis leads us to conclude that threshold levels impact resource use in our setting. Although we observe that the majority of subjects in each treatment extract at the assigned threshold level (Fig. 2), it is interesting to note that the average extractions do not always converge to the threshold level in all cases. For the strict threshold, subjects reduce extraction even when the threshold requirement is never met. While in the relaxed threshold, subjects reduce extraction well beyond the required threshold. In the moderate threshold, group extraction is around the threshold set.

It is interesting to note that the strict threshold treatment leads to a reduction in resource extraction despite the fact that the group target is never achieved. We also find that in the relaxed threshold treatment, groups reduce extraction beyond the level required to achieve the reward. We hypothesize that, apart from changing the subjects' incentives, the threshold level and the reward could have generated anchoring and learning effects that led to these behaviors. In our case, subjects could have used the assigned threshold level as their anchor and adjust their choices around that number in accordance with the "adjustment and anchoring" process (Tversky and Kahneman 1974). The threshold and reward might also create learning effects in which subjects learn that they can obtain higher payoffs when group members cooperate and reduce individual extraction from the common pool. This further induces players to try to cooperate, even when the group target is not met. Unfortunately, given our experimental design, it is not possible to identify the significance of these three reasons—incentives, anchoring, and learning effects—separately.

4.3 Impacts of varying collective thresholds on payoffs distribution and cost efficiency

Our results show that individual payoffs increase as rewards allow groups to achieve cooperation in the CPR setting. For all treatment groups, the payoffs from the treatment rounds are higher than the baseline rounds. This divergence can be clearly

¹⁴ In the baseline rounds, we cannot reject the null hypothesis that extraction levels are similar across groups (Kruskal–Wallis test statistic is 4.057). However, in the treatment rounds, the extraction levels across groups are significantly different at the 1 percent level (Kruskal–Wallis test statistic is 112.765).

¹⁵ Linear combination test yields a test statistic of -0.57 with p value of 0.57.

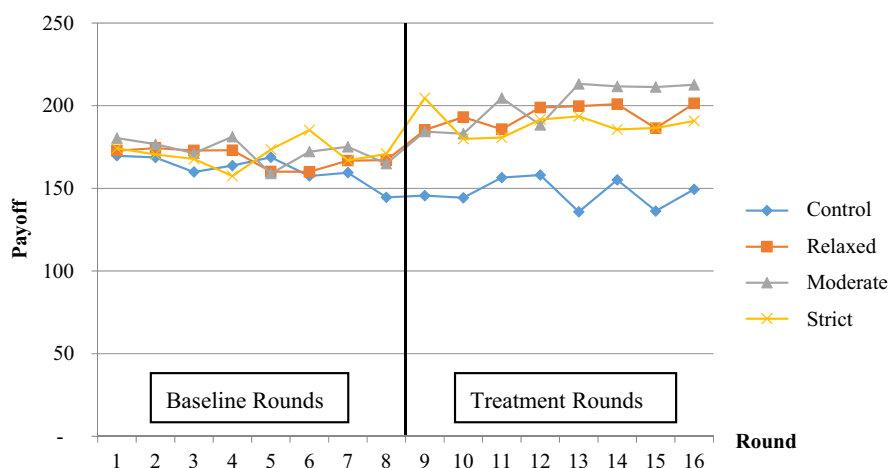


Fig. 3 Average individual payoff in each round

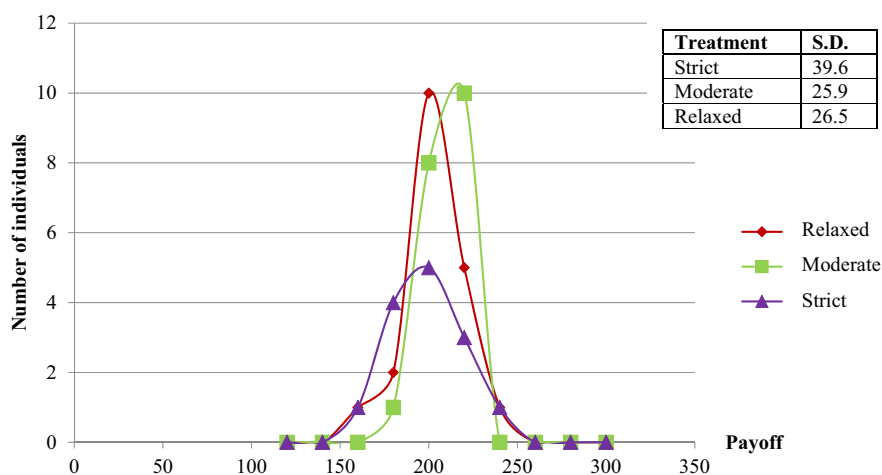


Fig. 4 Frequency distribution of payoffs under different threshold treatments

seen from Fig. 3. This finding suggests that, in addition to potential conservation gains, the use of rewards could also help increase incomes for users of the common resource provided that they are able to extract some resource for their own personal use. Applying the Kruskal Wallis test to the data on payoffs in the treatment rounds gives a test statistic of 150.70 with a p value of 0.0001. This shows that the three levels of collective thresholds have different effects on payoffs. Figure 3 shows that the moderate threshold generates the highest payoffs on average, followed by the relaxed and strict thresholds respectively.

Comparing the distribution of payoffs from the three treatments show that there are differences (see Fig. 4). The moderate threshold results in a more equitable

Table 4 Efficiency per Baht of reward payment

Treatment	Average group extraction		Average reduction in resource extraction	Average reward payment (Baht/group)	Average reduction per 1 Baht
	Rounds 1–8	Rounds 9–16			
Relaxed threshold	223	191	33	87.50	0.37
Moderate threshold_20	207	139	68	56.25	1.20
Moderate threshold_40	230	131	99	125.00	0.79
Strict threshold	211	125	86	0.00	n.a

distribution. The relaxed threshold level also shares this pattern but with a slightly higher dispersion. The strict threshold treatment results in the highest variance in payoffs. We believe this happens because, under the strict threshold treatment, subjects realize that meeting the requirement is almost impossible. Given that common belief, subjects' behaviors diverge into two groups. While most subjects try to cooperate by extracting zero, some subjects decide to extract a lot of resource. Cooperators who extract less receive little payoff in return for their cooperation, while non-cooperators free-ride on the cooperators' efforts to reap significant gains for themselves. In the moderate and relaxed thresholds, subjects are able to achieve the target some of the time. This experience is likely to lead to a belief that they can achieve the required target and obtain the reward if they cooperate. As a result, individual extractions are more clustered around the threshold level (Fig. 2), and individual payoffs are more equally distributed. Thus, in terms of equity in income distribution under various reward thresholds, using moderate and relaxed thresholds are better than the strict threshold. These results highlight the advantages of the moderate and relaxed thresholds especially on the social aspect.

For the strict threshold treatment, payoffs with treatment are higher than the baseline despite the fact that none of the groups were able to meet the target of no extraction so they did not have added payoffs from the reward. Thus, we believe that the higher payoff for the strict threshold case comes from the ability of the reward treatment to move people more towards the social optimum outcome via the anchoring and learning effects discussed earlier. Measuring cost-efficiency by extraction reduction per unit of reward spent, we find that the strict threshold is the most cost-efficient option since no reward payment is required to achieve the reduction in resource use. The moderate threshold performs second best in this sense. The relaxed threshold performs worst in terms of the average reduction per 1 Baht of reward. (Table 4).

4.4 Impacts of reward size on cooperation and resource use

To study the impact of reward size, we varied the reward amount to be low ($R = 20$) and high ($R = 40$) in the moderate threshold treatment. For both reward levels, the NE of extraction is at 80. Thus, a priori, we do not expect reward size to impact extraction decision. We find that average extraction in the low reward treatment is

Table 5 Regression results [Models (3) and (4)]

	(3) RE		(4) FE	
	Coefficient	SE ^a	Coefficient	SE ^a
Moderate threshold_20_SR	– 13.258***	2.475	– 12.532***	2.570
Moderate threshold_40_SR	– 21.563***	3.731	– 25.052***	3.503
Moderate threshold_20_LR	– 24.855***	4.146	– 26.386***	3.756
Moderate threshold_40_LR	– 24.024***	4.372	– 26.365***	4.551
Strict threshold SR & LR	Yes		Yes	
Relaxed threshold SR & LR	Yes		Yes	
Group fixed effects	Yes		No	
Session fixed effects	Yes		Yes	
Socio-economic controls	Yes		No	
Wald Chi-square	130.79***		120.19***	
Observations	1275		1360	
Subjects	85		85	
Time periods	15		15	

The baseline is the control group with no treatments

***, **, and * next to the coefficient indicate significance levels of 1 percent, 5 percent, and 10 percent accordingly

^aReported figures are bootstrapped standard errors

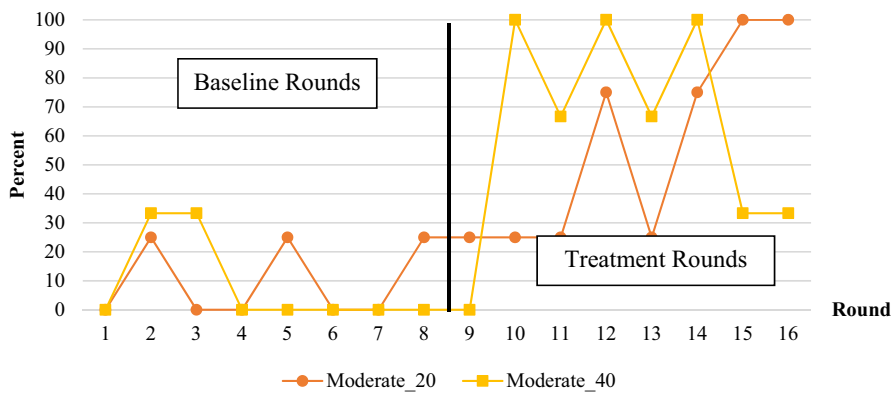


Fig. 5 Percentage of groups that reach the collective target in each round

41.3 in the baseline rounds and is 27.8 in the treatment rounds. For the high reward treatment, extraction is 45.9 in the baseline rounds and is 26.2 in the treatment rounds. The Wilcoxon signed-rank sum tests show that both treatments are effective in reducing resource use (test statistics = 3.061 and 3.459 for low and high rewards). However, the Kruskal–Wallis test shows that there are no significant differences in extraction choices between the two treatments (test statistic = 0.44, p value = 0.5). Regression analysis confirms these results (Table 5). The post-estimation test does

not show significant differences between the coefficients of the high and low reward treatments (test statistic = -0.23 , p value of 0.82).¹⁶

The key difference between high and low reward sizes is the speed with which the group targets are achieved. At the high reward size, the group target is achieved from the first round the reward is introduced (Fig. 5). Once the target is reached, individuals tend to maintain the group target in subsequent rounds although not all groups achieve the target all the time. At the low reward size, however, it takes three rounds after receiving the treatment for some groups to reach the target.

Regression analysis in Table 5 confirms the above findings. We examine this by splitting the data into the short run (Rounds 9–12) and the long run (Rounds 13–16) with the former indicating the periods immediately after the first application of the reward treatment and the latter the later periods where subjects have some experience with the rule. Models (3) and (4) show that the high reward case lowers extraction faster than the low reward case. In the FE specification, the coefficient of moderate threshold in the short run (labelled as moderate threshold_20SR in Table 5) is -12.532 and the coefficient in the long run (moderate threshold_20LR) is -26.386 . By contrast, at the high reward size, the coefficient is -25.052 in the short run and is -26.365 in the long run. Post-estimation tests of differences in coefficients rejects the null hypothesis of coefficient equality for the high and low reward treatments only in the short run (test statistic = 2.5 , p value = 0.012 in the short run) not the long run. This means that increasing reward size is less cost-efficient (Table 4). In practice, when budgets are limited, using a low reward size could help achieve the goal without breaking the bank.

5 Conclusions and recommendations

This paper studies the topic of institutional design when rewards are used to induce cooperation in a common pool resource setting. We look at whether or not rewards are effective in limiting resource use and examine how differences in collective threshold levels to obtain the rewards compare on (i) resource extraction, (ii) pay-offs distribution and (iii) cost efficiency. We also look at the impact of reward size. We find that, in our setting, using rewards is associated with positive conservation outcomes, and that its impacts depend on the design of the reward instrument. Our main findings can be summarized as follows;

Using rewards is positively associated with conservation gains in our common pool resource setting. Consistent with the existing literature, our findings indicate that the application of rewards is effective in reducing resource extraction in a social dilemma situation.

In our setting, we find that the strict and moderate thresholds are good at reducing resource use but differ in terms of cost efficiency and payoffs distribution. The design of the reward instrument in terms of target setting matters in obtaining desirable outcomes. Overall, of the three threshold levels, the strict and moderate levels

¹⁶ The test applied is a linear combination test.

result in the largest amount of extraction reduction. One advantage of the strict threshold is that it is most cost-efficient. However, it induces free-riding behavior in group members, leading to a payoffs distribution with the highest variance. The moderate threshold, on the other hand, is found to be as effective in limiting resource use, but with a more equitable distribution of payoffs. Thus, if equity is a concern, applying rewards with moderate threshold levels might be a more appropriate option. The relaxed threshold treatment is not found to significantly reduce resource use.

Based on our findings, we hypothesize that rewards work through providing economic incentives, and by generating anchoring and learning effects. Rewards push people towards the social optimum by enticing them with a prize for reducing extraction (i.e. providing economic incentives). Given that the strict threshold treatment groups tried to reduce extraction even though their target is never achieved, it could be that the threshold level could serve as an anchor around which subjects adjust their extraction levels. The reward treatments further allow people to learn that their payoffs increase if they conserve resources. This encourages them to try to reduce extraction even though they may not be able to obtain the reward. The learning effect is clearly illustrated in the strict threshold case where the average extraction with treatment is lower even though none of the groups are able to meet the target, and in the relaxed and moderate threshold cases since players ultimately reduce extraction to below the level required to achieve the rewards.

Our analysis suggests that reward size does not affect long-run resource extraction, but changes the speed of realizing the equilibrium outcome. Keeping the threshold constant, we find that changing the reward size does not significantly impact resource extraction in the long term, but the higher reward does spur groups to try to meet the target faster. Because reduction in resource extraction is similar, but reward payments are different, the lower reward size is more cost-efficient.

Finally, in a paper that studies the application of monetary rewards such as ours, the issue of motivational ‘crowding out’ whereby the application of reward decreases intrinsic motivation to conserve the common pool resource is an issue of concern. In the context of positive incentives applied to forest resources, the finding in the literature is mixed. While Midler et al. (2015) and Narloch et al. (2012) find evidence for motivational crowding out, Andersson et al. (2018) and Salk et al. (2017) find no evidence for it. In our case, the result showing that groups still reduced resource use even though they were not able to obtain the reward provides some indication that motivational crowding out might not be a problem. Furthermore, Andersson et al. (2018)’s finding that social conditions such as communication and trust can help sustain conservation in the long run, and Kaczan et al. (2019)’s finding that socio-demographic and farm characteristics could predict motivational crowding out or in suggest that rewards have a better chance of working in some contexts and sub-populations more than others. The specific contexts and sub-populations where different levels of collective thresholds could sustain conservation in the long run will need to be further studied.

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